

Sec 3.5 – The Trigonometric Functions

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

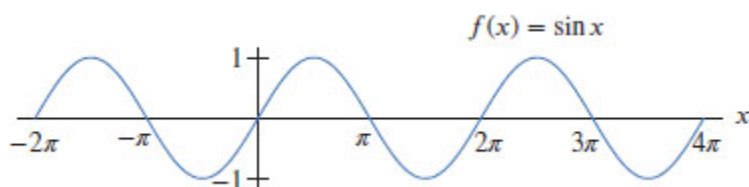
What does it mean for a function to be **periodic**?

When the values of a function repeat.

For a periodic function of time, we say that the **period** is the smallest time needed for the function to complete one cycle.

The **amplitude** is half the distance between the maximum and minimum values, if they exist.

The sine and cosine functions are periodic. So, we would expect their derivatives to be periodic, since they are derivative functions. Note that all our work in this class with trigonometric functions is using **radians**, and not degrees.



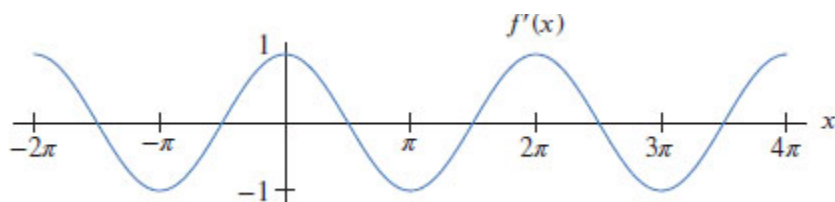
Where is $\sin(x) = 0$? At $x = 0, \pm\pi, \pm2\pi \pm 3\pi, \dots$

Where is the derivative of $\sin(x)$, $\frac{d}{dx}\sin(x) = 0$? At $x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \dots$

Where are the largest ("most positive") slopes of $\sin(x)$? At $x = 0, \pm2\pi \pm 4\pi, \dots$

Where are the smallest ("most negative") slopes of $\sin(x)$? At $x = \pm\pi \pm 3\pi, \dots$

Based on this information, we can graph $\frac{d}{dx}\sin(x)$:



This looks suspiciously like $\cos(x)$.

Let's consider the limit of the difference quotient, or derivative, of $\sin(x)$ at $x = 0$.

$$\lim_{h \rightarrow 0} \frac{\sin(0 + h) - \sin 0}{h} = \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

Partner up

Numerically estimate this limit with successively smaller values of h :

h (radians)	-0.1	-0.01	-0.001	-0.0001	0.0001	0.001	0.01	0.1
$(\sin h)/h$	0.99833	0.99998	1.0000	1.0000	1.0000	1.0000	0.99998	0.99833

We can do the above for $\cos(x)$ as well. This leads to the following theorem:

Theorem

For x in radians,

$$\frac{d}{dx}(\sin x) = \cos x \quad \text{and} \quad \frac{d}{dx}(\cos x) = -\sin x.$$

As practice, using the above as well as the rules we've covered in Chap 3 so far, calculate the following derivatives:

$$y = 3x^2 + 3 \cos(x) - 4 \sin(x) \quad \frac{dy}{dx} = 6x - 3 \sin(x) - 4 \cos(x)$$

$$\begin{aligned} \frac{dy}{dx} &= \left(\frac{d}{dx} \sin(x) \right) \cos(x) + \sin(x) \left(\frac{d}{dx} \cos(x) \right) \\ &= (\cos(x)) \cos(x) + \sin(x)(-\sin(x)) \\ &= \cos^2(x) - \sin^2(x) \end{aligned}$$

$$y = \sin(x) \cos(x)$$

Partner up

1. $y = \tan(x) = \frac{\sin(x)}{\cos(x)}$
2. $y = \sec(\theta) = \frac{1}{\cos(\theta)}$
3. $y = \cos^3(x) = (\cos(x))^3$
4. $y = \tan(t^3)$
5. $y = \cos(3\theta)$
6. $y = e^{\sin(x^2)}$

1. $y = \tan(x) = \frac{\sin(x)}{\cos(x)}$

Solution:
$$\begin{aligned} y' &= \left(\frac{\sin(x)}{\cos(x)} \right)' = \frac{\cos(x)(\sin(x))' - \sin(x)(\cos(x))'}{\cos^2(x)} \\ &= \frac{\cos(x) \cos(x) - \sin(x)(-\sin(x))}{\cos^2(x)} = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} \\ &= \frac{1}{\cos^2(x)} (= \sec^2(x)) \end{aligned}$$

2. $y = \sec(\theta) = \frac{1}{\cos(\theta)}$

Solution:
$$\begin{aligned} y' &= \left(\frac{1}{\cos(x)} \right)' = \frac{\cos(x)(1)' - 1(\cos(x))'}{\cos^2(x)} = \frac{0 - (-\sin(x))}{\cos^2(x)} \\ &= \frac{\sin(x)}{\cos^2(x)} = \frac{1}{\cos(x)} \frac{\sin(x)}{\cos(x)} = \frac{\tan(x)}{\cos x} (= \sec x \tan x) \end{aligned}$$

3. $y = \cos^3(x) = (\cos(x))^3$

Solution: Using the chain rule:

$$y' = 3(\cos(x))^2(-\sin(x)) = -3\cos^2(x)\sin(x)$$

4. $y = \tan(t^3)$

Solution:

$$y' = \frac{1}{\cos^2(t^3)}(t^3)' = \frac{3t^2}{\cos^2(t^3)}$$

5. $y = \cos(3\theta)$

Solution: $y' = -\sin(3\theta)(3\theta)' = -\sin(3\theta)(3) = -3\sin(3\theta)$

6. $y = e^{\sin(x^2)}$

Solution:

$$y' = e^{\sin(x^2)}(\sin(x^2))' = e^{\sin(x^2)}(\cos(x^2))(x^2)' = e^{\sin(x^2)}(\cos(x^2))2x$$

These derivative rules should be memorized:

Theorem

For x in radians,

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \frac{1}{\cos^2 x}$$

"Just in time" Trig review!

Reciprocal identities:

$$\csc(\theta) = \frac{1}{\sin(\theta)}$$

$$\sec(\theta) = \frac{1}{\cos(\theta)}$$

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

Given $\sin(Bt)$ or $\cos(Bt)$, the **period**, or length of the cycle, is $\frac{2\pi}{|B|}$.

Pythagorean identities:

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

$$1 + \cot^2(\theta) = \csc^2(\theta)$$

$$\tan^2(\theta) + 1 = \sec^2(\theta)$$

Sum identities:

$$\sin(a + b) = \sin(a)\cos(b) + \sin(b)\cos(a)$$

$$\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$$

Partner up

With t in years, the population of a herd of deer is represented by

$$P(t) = 4000 + 500 \sin \left(2\pi t - \frac{\pi}{2} \right).$$

1. How does this population vary with time?
2. When in the year is the population a maximum? What is that maximum? Is there a minimum? If so, when?
3. When is the population growing fastest? When is it decreasing fastest?
4. How fast is the population changing on July 1?

1. $\sin()$ varies between -1 and 1. So the max from the second term will be +500, and the min will be -500. So $P(t)$ will vary between 4000-500 and 4000+500, or between 3500 and 4500.
2. $P(t)$ is maximum, 4500, when $\sin \left(2\pi t - \frac{\pi}{2} \right) = 1$, or when $2\pi t - \frac{\pi}{2} = \frac{\pi}{2}$, $2\pi t = \pi$, $t = \frac{1}{2}$.
 $P(t)$ is minimum, 3500, when $\sin \left(2\pi t - \frac{\pi}{2} \right) = -1$, or when $2\pi t - \frac{\pi}{2} = -\frac{\pi}{2}$, $2\pi t = 0$, $t = 0$.
3. The population is growing fastest when the derivative of $P(t)$, $P'(t)$ or $\frac{dP}{dt}$, is biggest.
Since $P'(t) = 500 \cos \left(2\pi t - \frac{\pi}{2} \right) \cdot 2\pi$, this is largest when $\cos \left(2\pi t - \frac{\pi}{2} \right) = 1$, or $2\pi t - \frac{\pi}{2} = 0$, $t = \frac{1}{4}$.
 $P'(t)$ is smallest when $\cos \left(2\pi t - \frac{\pi}{2} \right) = -1$, or $2\pi t - \frac{\pi}{2} = \pi$, $t = \frac{3}{4}$.
4. Since t is in years, July 1 is halfway through the year, or $\frac{1}{2}$. We simply plug this into the derivative, to find the rate of change on July 1.
 $P' \left(\frac{1}{2} \right) = 500 \cos \left(2\pi \frac{1}{2} - \frac{\pi}{2} \right) \cdot 2\pi = 500 \cos \left(\frac{\pi}{2} \right) \cdot 2\pi = 0$.
This means that when the population is a maximum (**see #2**), the rate of change of the population is 0.